

ESP Study

MATH 213

Group 1

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

Brennan: 12

Easton: 14

Brodey: 14

A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing?

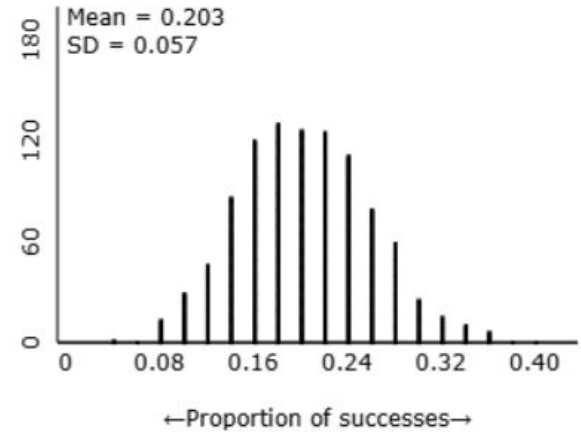
According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?

All of our group members were psi hitting, with Easton and Brodey being further away than the expected number of 10, both having 14 hits.

Use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the ESP test according to the **null model** proposed by the skeptic, where you are just guessing. To set this up, think about what the `probability of success` should be for this model. (It is no longer 0.5!)

Paste a picture of your histogram of correct predictions for the null model here. Describe the distribution. As usual, give a W/W/W/W sentence, address the center, shape, and spread of the distribution, and speak in context.

The histogram shows 1000 simulations of the ESP test where each test recorded hits out of 50 attempts. While assuming that the null hypothesis is true, the mean value of our graph is 0.203, showing a normal distribution, with a standard deviation of 0.057.



Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psychic group member's results?

Group 2

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

Colin: 11

Sophia: 7

A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing?

10

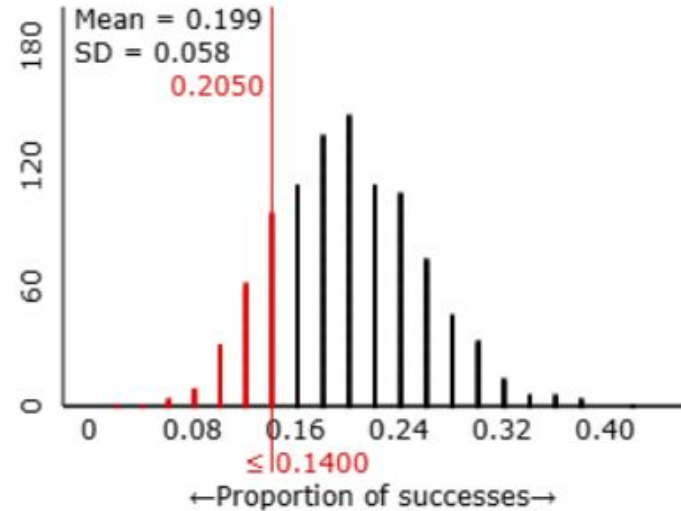
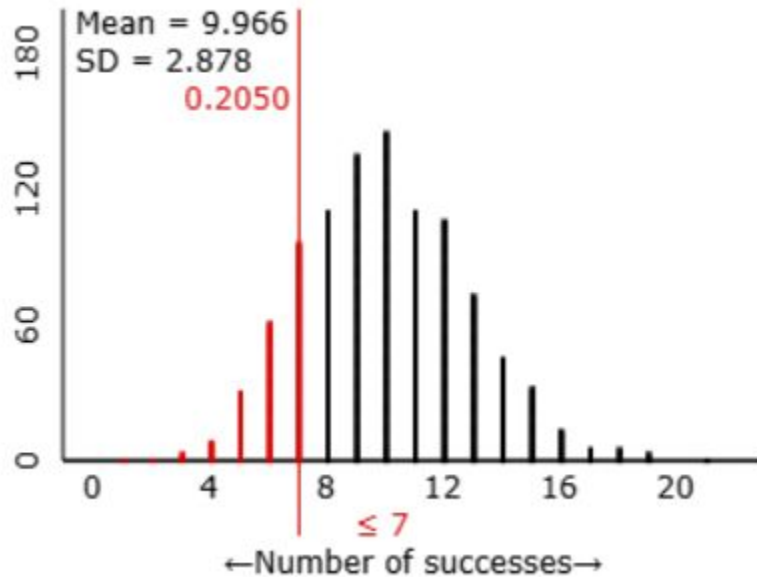
According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?

Sophia showed more evidence of ESP. Since she is 3 away from the expected number and Colin is 1 away from the expected number Sophia had more evidence of psi missing than Colin did of psi hitting.

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Paste a picture of your histogram of correct predictions for the null model here. Describe the distribution. As usual, give a W/W/W/W sentence, address the center, shape, and spread of the distribution, and speak in context.

The histogram shows 1000 simulations of the ESP test under the assumption of the null hypothesis, which is the guesses are random. The histogram is centered around 0.2 with a right skew and a standard deviation of 0.058.



Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psychic group member's results?

The p-value for our most extreme result is 0.205. Yes the null model does seem like a plausible explanation for the results because the probability of picking 7/50 correctly is approximately 20% making it likely that this was based on chance.

Group 3

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing?

According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?

Use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the ESP test according to the **null model** proposed by the skeptic, where you are just guessing. To set this up, think about what the `probability of success` should be for this model. (It is no longer 0.5!)

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Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psychic group member's results?

Group 4

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

Emily: 12

Itunu: 14

A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing?

The skeptic would expect 10 correct cards predicted out of 50 total cards.

According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?

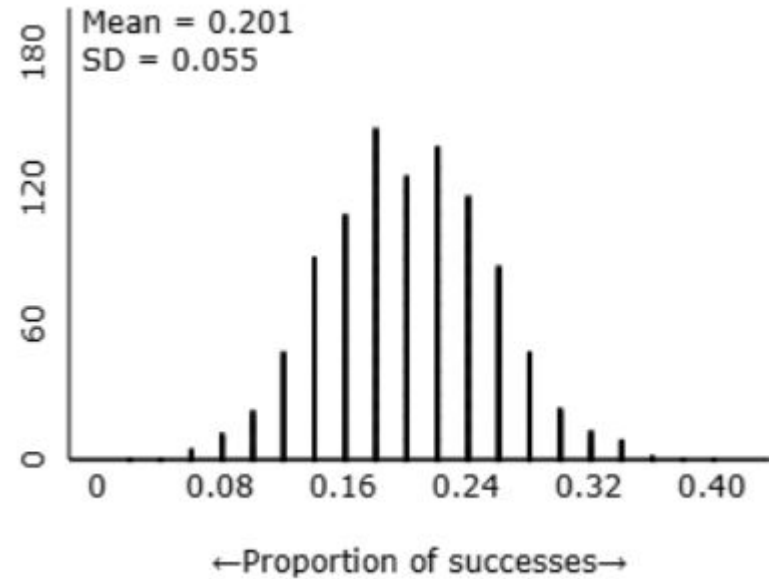
No, we got greater than 10 but not by a significant amount. Itunu was farther.

Use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the ESP test according to the **null model** proposed by the skeptic, where you are just guessing. To set this up, think about what the `probability of success` should be for this model. (It is no longer 0.5!)

Paste a picture of your histogram of correct predictions for the null model here. Describe the distribution. As usual, give a W/W/W/W sentence, address the center, shape, and spread of the distribution, and speak in context.

The dotplot describes the proportion of correctly predicted cards out of 50 total cards for each of the 1000 simulated experiments.

The dotplot has a normal distribution. The typical proportion of correct cards is .201, give or take .055.

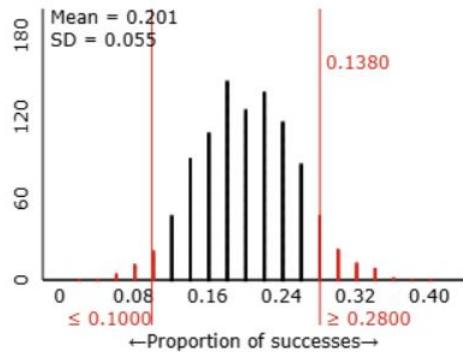


Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psyhic group member's results?

.138

Yes, because in our simulated experiments, the proportion that is as or more extreme than our result was not too small. That is, it not unusual to get this result from guessing.



Group 5

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

Jonas: 12 cards out of 10

A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing?

About $\frac{1}{5}$ or 10 cards out of 50

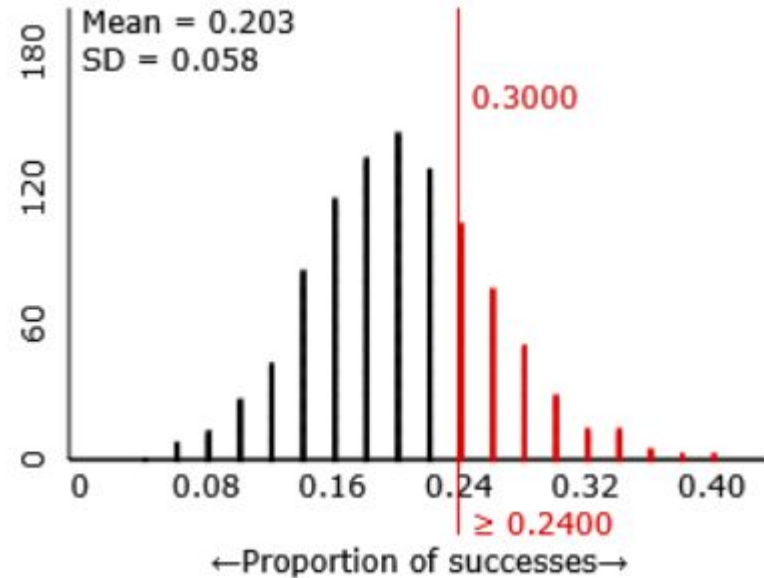
According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?

My results were higher than the expected proportion of 0.2, being 0.24.

Use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the ESP test according to the **null model** proposed by the skeptic, where you are just guessing. To set this up, think about what the `probability of success` should be for this model. (It is no longer 0.5!)

Paste a picture of your histogram of correct predictions for the null model here. Describe the distribution. As usual, give a W/W/W/W sentence, address the center, shape, and spread of the distribution, and speak in context.

This graph represents the resulting proportions from a randomized simulation of our experiment under the null hypothesis. The graph is normal, with a mean value of proportions of correctly chosen cards being 0.203 with a standard deviation of 0.058. Most of our data is roughly between 0.14 and 0.26.



Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

P-value = 0.3

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psychic group member's results?

The proportion of correct cards from our test statistic of 0.24 falls within the reasonable range of probabilities, having a p-value of 0.3. It seems very likely that our results were because of random chance, so we will retain the null hypothesis.

Group 6

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

Stone - I guessed 6 cards correctly.

Jonathan - 11 out of 50 and I picked the waves every time.

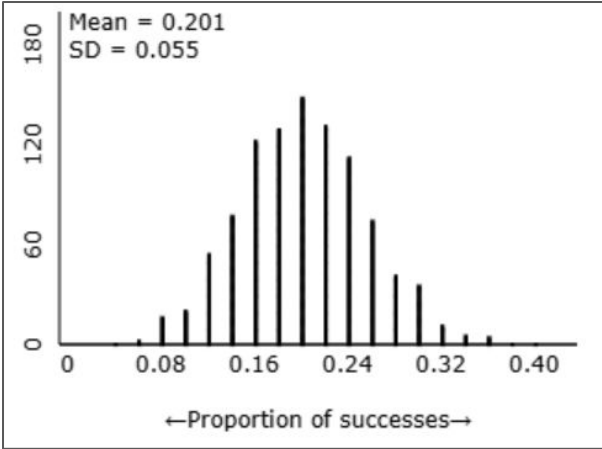
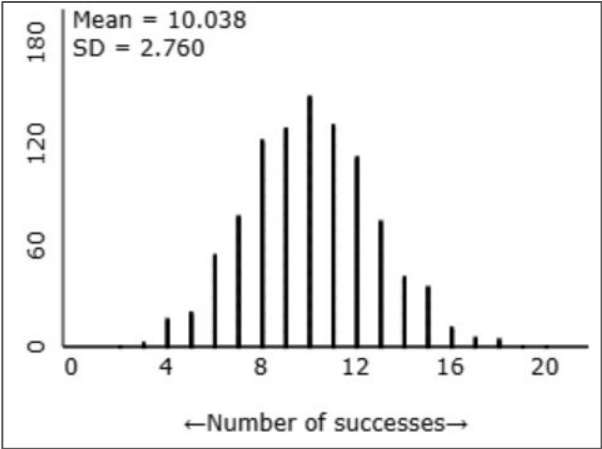
A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing?

There is only a 1 out of 5 chance to get a choice correct, so they would assume about 10 guesses would be correct out of 50 guesses.

According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?

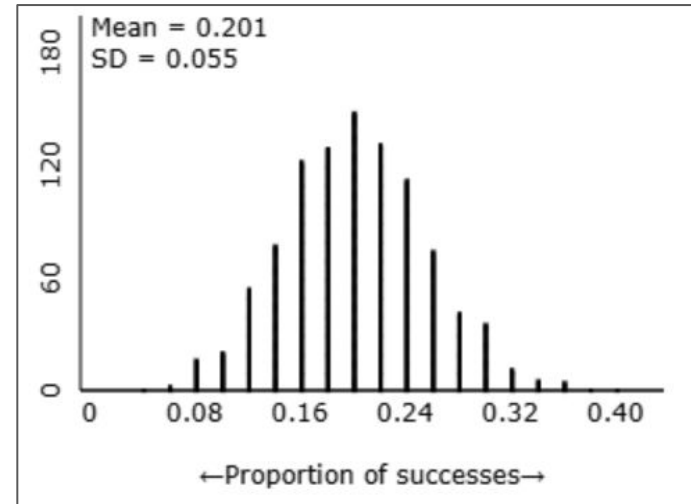
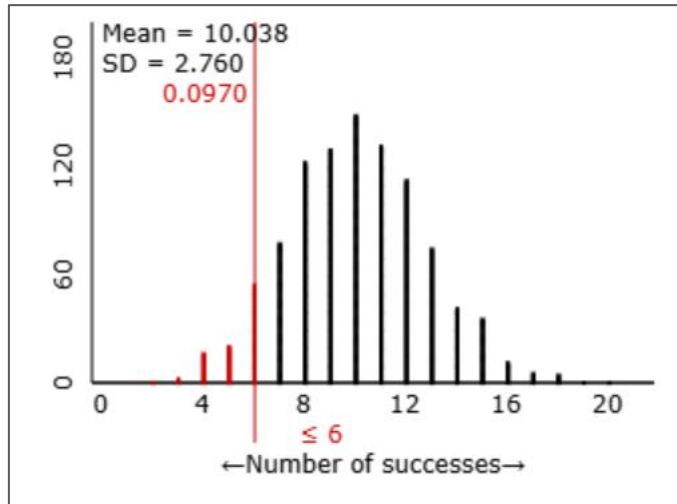
No, I don't think that anyone shows evidence of psi hitting or missing. Stone was the furthest away from the expected number with only 6 correct guesses out of 50.

Use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the ESP test according to the **null model** proposed by the skeptic, where you are just guessing. To set this up, think about what the 'probability of success' should be for this model. (It is no longer 0.5!)



Paste a picture of your histogram of correct predictions for the null model here. Describe the distribution. As usual, give a W/W/W/W sentence, address the center, shape, and spread of the distribution, and speak in context.

The histogram shows a normal bell curve distribution and is centered at 0.201 (proportion graph). A mean of 10.038 (counts) shows that the average number of successes in a 50 card deck is about 10. Stone guessed 6 cards correctly, and in our simulated runs, we had 97/1000 runs in which we achieved the same or similar results. Assuming the null hypothesis is true, the odds that Stone getting his result is about 9.7%. This means that we cannot reject the null hypothesis, and further testing would be required for affirmation of the alternative hypothesis.



Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

0.0970

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psychic group member's results?

[See prior page where we discussed in summary.](#)

Group 7

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

Moyed: 11

Daniel: 7

A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing?

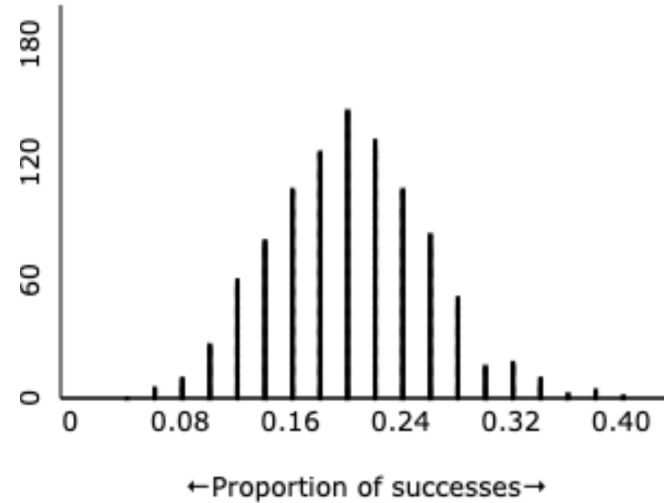
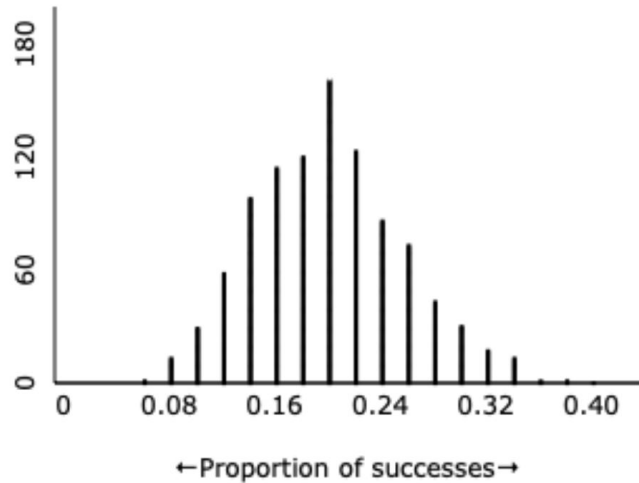
10

According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?

Daniel further away with 7

Use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the ESP test according to the **null model** proposed by the skeptic, where you are just guessing. To set this up, think about what the `probability of success` should be for this model. (It is no longer 0.5!)

Paste a picture of your histogram of correct predictions for the null model here. Describe the distribution. As usual, give a W/W/W/W sentence, address the center, shape, and spread of the distribution, and speak in context.



The graph has a normal bell shaped distribution, with a range of 4% to 36% of cards guessed right. The average percent of cards guessed right was 20%

Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

$P = 0.853$

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psychic group member's results?

$P = 0.853$.

Group 8

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing?

According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?

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Paste a picture of your histogram of correct predictions for the null model here. Describe the distribution. As usual, give a W/W/W/W sentence, address the center, shape, and spread of the distribution, and speak in context.

Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psychic group member's results?

Group 9

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

Alexis: 10

Landon: 10

A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing?

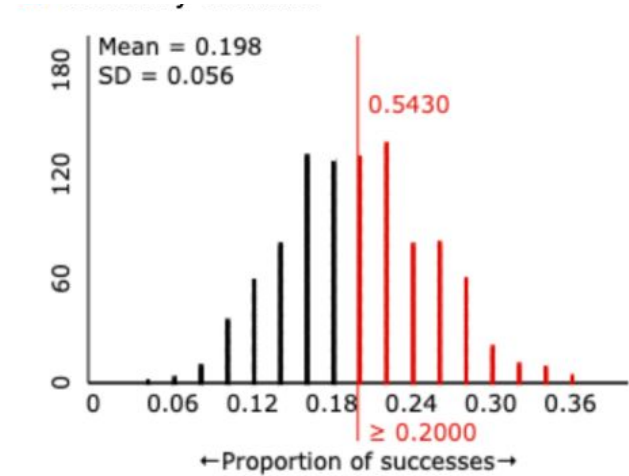
I would say 10 because there are 5 cards to choose from, and it's reasonable to assume that you'd get at least 1 or two right per ten cards

According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?
Nope.

Use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the ESP test according to the **null model** proposed by the skeptic, where you are just guessing. To set this up, think about what the `probability of success` should be for this model. (It is no longer 0.5!)

Paste a picture of your histogram of correct predictions for the null model here. Describe the distribution. As usual, give a W/W/W/W sentence, address the center, shape, and spread of the distribution, and speak in context.

This graph measures the count of proportions in which someone will correctly guess out of a selection of 5-incorrectly guess assuming the null hypothesis is true. The graph is in the shape of a normal bell curve with an average distance of .056 between points. The amount of points that are greater than or equal to .2 is 54%, meaning that it's very likely that the null hypothesis is accepted.



Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psychic group member's results?

Our proportions should both be .2, so I'd say, definitely

Group 10

A standard tool in testing for extra-sensory perception (ESP) is a deck of "Zener cards." Each card has one of five symbols: a five-pointed star, three wavy lines, a circle, a plus symbol, or a square.

In one ESP test, a subject is asked to predict the symbol on 50 randomly chosen Zener cards. Try this test yourself (**choosing a 50-card trial**) at the following web site:

<https://psychicscience.org/esp3>

How many cards did you guess correctly? Each person in the group should respond by name.

Jonah: 15

Jose: 5

A skeptic who does not think that you have ESP would suggest that your result could just have come from guessing. How many of the 50 cards would the skeptic expect you to predict correctly, assuming that you were just guessing? I think you would predict correctly 10 because there is 50 chances and 5 cards each chance.

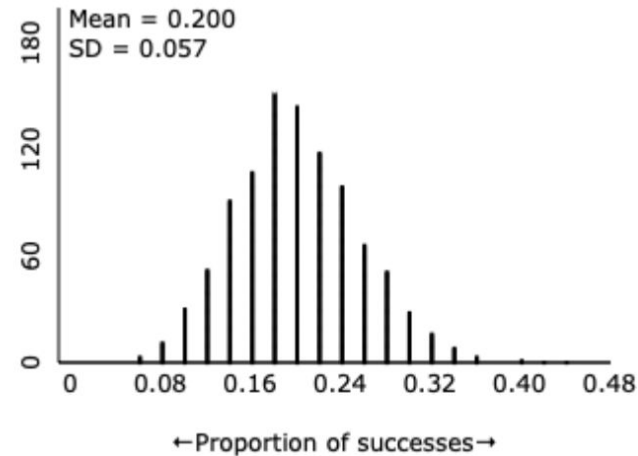
According to the testing web site, both predicting **more** cards than expected ("psi hitting") and **fewer** cards than expected ("psi missing") are evidence of ESP. Do any of your group members' results show evidence of psi hitting or of psi missing? Which member was further away from the expected number?

For the study Jonah was psi hitting with guessing 15 right and Jose psi missing with only guessing 5 right.

Use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the ESP test according to the **null model** proposed by the skeptic, where you are just guessing. To set this up, think about what the `probability of success` should be for this model. (It is no longer 0.5!)

Paste a picture of your histogram of correct predictions for the null model here. Describe the distribution. As usual, give a W/W/W/W sentence, address the center, shape, and spread of the distribution, and speak in context.

In the data we have there is a ESP test. The number of rounds we had gone was 50, and the number of guesses for each instance was 5. We had 1000 simulations to see if 10 out of 50 was odd. The data is bell shaped and shows that it is centered around 0.2. In the 1000 simulations we had were in between the points 0.075 and 0.35, with most of the points being in between 0.15 and 0.28.



Use the applet to estimate a p-value for your group's most extreme result under the null model. This p-value is the probability of seeing data like this result or "more extreme" assuming that you are guessing cards at random, as described precisely by the null model.

Report your p-value here. Does the null model seem like a plausible explanation for your apparently-most-psychic group member's results?

The p-value that we had for our graph was .06300 which suggest 63 correct out of the 1000 simulations.